

Business Insight Report

What Drives Purchasing Behaviour in E-Commerce

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Executive Summary

This report analyses 9,994 e-commerce transactions to identify what drives — and what quietly destroys — profitability across customer segments, regions, product categories, and discounting practices. While headline revenue figures appear healthy, the analysis reveals that discounting behaviour is the single largest controllable risk to the business, actively converting a meaningful share of orders into net losses. The report also identifies specific underperforming product sub-categories, a churn risk building among previously valuable customers, and a mismatch between revenue and profitability at the individual customer level.

Metric	Value	Insight
Total Revenue	₹22.97L	Baseline scale of business
Total Orders	9,994	Full dataset analysed
Overall Profit Margin	~12.5%	Masks severe internal variance
Worst Discount Band Margin	-77.40%	933 orders actively lose money

Methodology

Data was cleaned in Excel, loaded into a MySQL database, and analysed using SQL to answer seven structured business questions covering segment, regional, category, discount, customer, and seasonal performance. A weighted profit margin measure ($\text{Total Profit} \div \text{Total Sales}$) was used throughout rather than a simple row-level average, to avoid distortion from outlier transactions. Findings were visualized in a four-page interactive Power BI dashboard incorporating RFM (Recency, Frequency, Monetary) customer segmentation.

Key Findings & Recommendations

Finding 1: Discounting Is Actively Destroying Profit

Orders with no discount achieve a healthy 29.51% profit margin. Once discounts exceed 40%, margin collapses to -77.40% — meaning the business loses more than

it earns on every such sale. This is not an isolated issue: 933 orders (9.3% of all transactions) fall into this high-discount band.

Recommendation: *Implement a discount approval workflow requiring managerial sign-off for any discount above 20%, with a hard cap or additional review at 40%+.*

Finding 2: Revenue and Profitability Are Not the Same Thing

Sean Miller, the single highest-revenue customer, is unprofitable overall. A ₹22,638 order discounted 50% resulted in a ₹1,811 loss on that transaction alone, dragging his account into negative profit despite top-line revenue leadership.

Recommendation: *Introduce a profit-per-account view (not just revenue-per-account) into sales reporting, and review discount authority on large individual deals before they are closed.*

Finding 3: Furniture Category Is a Profitability Drag

Furniture generates revenue comparable to Technology (₹7.4L vs ₹8.4L) but returns only 2.49% margin versus 17%+ elsewhere. The cause is concentrated in two sub-categories — Tables (-8.56% margin) and Bookcases (-3.02% margin) — both carrying the highest average discounts in the category (26.13% and 21.11% respectively).

Recommendation: *Reassess pricing and discount thresholds specifically for Tables and Bookcases; consider whether these sub-categories should carry lower discount ceilings than the rest of the catalogue.*

Finding 4: Central Region Underperforms Despite Strong Revenue

Central generates substantial revenue but posts the weakest profit margin of any region — nearly half that of West — suggesting excessive discounting, higher costs, or an unprofitable product mix specific to that region.

Recommendation: *Conduct a region-level discount and cost audit for Central to identify the specific driver before applying company-wide fixes.*

Finding 5: Customer Attrition Risk Is Building

RFM segmentation shows 178 customers already 'Lost' and a further 179 'At Risk' — together representing more accounts than the 174 'Champions' currently driving the most value. At-Risk customers alone represent ₹5,25,096 in historical revenue.

Recommendation: Launch a targeted re-engagement campaign (personalized offers, check-in outreach) for the At-Risk segment before further migration to Lost status.

Finding 6: Sales Are Highly Seasonal

November, December, and September account for significantly higher revenue than other months, consistent with holiday shopping and fiscal year-end purchasing. January and February fall to roughly 25-30% of peak-month revenue.

Recommendation: Align inventory, staffing, and marketing spend to the September-December peak; use January-February promotions to smooth the post-holiday slowdown rather than treating it as an off-season to ignore.

Conclusion

The business is not short of revenue — it is short of pricing discipline. The single highest-leverage action identified in this analysis is tightening control over discount authorization, since this one factor is directly implicated in the Furniture category loss, the Sean Miller account loss, and a 933-order segment of the business currently operating at a loss on every sale. Addressing this, alongside targeted attention to the Central region and the At-Risk customer segment, represents the clearest path to improving overall profitability without needing to grow revenue at all.